Mississippi Valley State University

# CS 341 — Discrete Structures

Practice Set 11

*Sets, Subsets, Power Sets, and Cartesian Products*

Covers Chartrand & Zhang, Sections 2.1 and 2.3

**Status of this set**

**Ungraded practice.** Solutions released with it. Attempt everything first.

# Part I: How to Practice This Material

## Count the outermost braces first

**Two habits that prevent most errors**

1. **Before answering anything about a set, list its elements at the outermost level and count them.** For $\{1, \{2, 3\}\}$ the answer is two: the number 1 and the set $\{2, 3\}$.
2. **$\in$ takes an element on the left; $\subseteq$ takes a set on the left.** If unsure, ask whether the thing on the left appears *in* the list, or is *made of* things in the list.

**Subset claims are quantified implications**

$A \subseteq B$ means $\forall x$, $(x \in A \Rightarrow x \in B)$. Proving it takes an arbitrary element; disproving it takes one element of $A$ outside $B$ — the same asymmetry as Wednesday.

**Three mistakes to expect**

1. **$|\{\emptyset\}| = 0$.** It is 1. A set containing the empty set is not empty.
2. **Confusing $\in$ and $\subseteq$**, especially when elements are themselves sets.
3. **$A \times B = B \times A$.** The pairs are ordered; only the sizes match.

# Part II: Practice Problems

## Section A — Element or subset?

**1.** Let $A = \{0, \{0\}, \{0,1\}\}$. Determine whether each is true or false, with a one-line reason.

(a) $0 \in A$

(b) $\{0\} \in A$

(c) $\{0\} \subseteq A$

(d) $\{\{0\}\} \subseteq A$

(e) $1 \in A$

(f) $\emptyset \subseteq A$

## Section B — Power sets

**2.** Write $\mathcal{P}(A)$ in full for each, and give $|\mathcal{P}(A)|$.

(a) $A = \{x, y, z\}$

(b) $A = \{\emptyset\}$

(c) $A = \{1, \{1\}\}$

**3.** Answer each with a number and a one-line reason.

(a) If $|A| = 6$, what is $|\mathcal{P}(A)|$?

(b) If $|\mathcal{P}(A)| = 128$, what is $|A|$?

(c) How many elements does $\mathcal{P}(\mathcal{P}(\emptyset))$ have?

## Section C — Cartesian products

**4.** Let $A = \{1, 2\}$ and $B = \{a, b\}$.

(a) List $A \times B$ and $B \times A$.

(b) Are they equal? Are their cardinalities equal?

(c) List $A \times A$.

(d) If $|A| = 5$ and $|B| = 4$, give $|A \times B|$ and $|\mathcal{P}(A \times B)|$.

## Section D — Writing

**5.** In four to six complete sentences, explain the difference between $\{0\}$, $\emptyset$, and $\{\emptyset\}$. Give the cardinality of each and say why confusing them breaks power-set problems.

## Section E — Challenge (optional)

**6.** Prove that $\emptyset \subseteq A$ for every set $A$, using the definition of subset as a quantified implication.